先进计算与数字工程研究中心

PATTERN RECOGNITION AND MACHINE LEARNING

Lecture 13 – Ensemble Learning I: Bagging, Random Forests, and the Bridge to Boosting



Chizhi Chris ZHANG

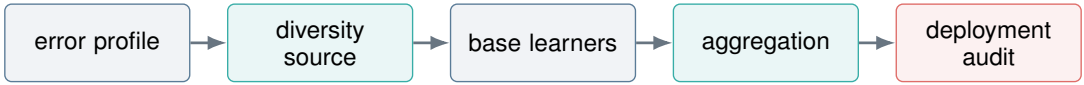
zhangchizhi@ciomp.ac.cn

Advanced Computing and Digital Technology Research Center

University of Chinese Academy of Sciences

Fall 2026

Average errors only when they are different



More learners help only until correlated error and compute dominate.

Guiding question

Which part of a learner's error can an ensemble actually remove?

- bagging perturbs training rows
- random forests also perturb split features
- averaging reduces independent variance
- correlated bias remains

Two hours of forests and one bridge to boosting

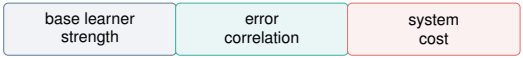
By the end

You can derive bootstrap coverage, implement OOB evaluation, tune random forests, interpret importance cautiously, and explain why boosting is a different strategy.

Period	Core content	Decision
1	bootstrap and bagging	how does averaging lower variance?
2	random forests and OOB	how much randomness is useful?
3	reliable tuning and boosting bridge	which ensemble matches the error?

Track strength, correlation, and cost together

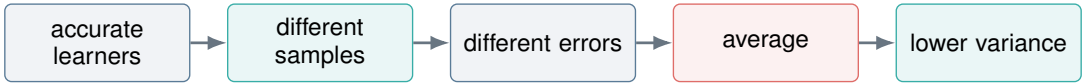
先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER



What to calculate

Every ensemble experiment records base performance, pairwise prediction correlation, aggregate performance, calibration, memory, and latency.

When does averaging many predictors reduce generalization error?



Averaging reduces the independent part of variance but leaves common error untouched.

Question

When does averaging many predictors reduce generalization error?

Example

unstable decision trees trained on nearby samples

Main idea

Averaging reduces the independent part of variance but leaves common error untouched.

Ensemble members need different errors

	测试例 1	测试例 2	测试例 3
f_1	✓	✓	×
f_2	×	✓	✓
f_3	✓	×	✓
集成	✓	✓	✓

（a）集成提升性能

	测试例 1	测试例 2	测试例 3
f_1	✓	✓	×
f_2	✓	✓	×
f_3	✓	✓	×
集成	✓	✓	×

（b）集成不起作用

	测试例 1	测试例 2	测试例 3
f_1	✓	×	×
f_2	×	✓	×
f_3	×	×	✓
集成	×	×	×

（c）集成起负作用

An ensemble helps when member errors differ; averaging identical errors does not help.

Ensemble diversity and error correlation: model and variables

Model and assumptions

An ensemble combines predictions from base learners whose individual errors are accurate enough and not perfectly correlated.

$$\mathrm{Var}(\bar{f}) = \rho\sigma^2 + \frac{1-\rho}{M}\sigma^2$$

Symbol	Meaning	Status
$\bar{f}$	ensemble mean	average prediction
ρ	pairwise error correlation	limits the variance floor
M	number of learners	reduces only independent variance

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Ensemble diversity and error correlation: worked example

Quantity	Value	Meaning
base variance	1.00	each learner
$M = 100, \rho = 0$	0.01	mostly removable
$M = 100, \rho = 0.6$	0.604	correlation dominates

What the result means

Increase diversity or change the base learner before increasing ensemble size further.

Ensemble diversity and error correlation: what to check

Main idea

Averaging reduces the independent part of variance but leaves common error untouched.

What to check

base-error correlation and ensemble variance

Limit

adding nearly identical learners increases cost without meaningfully lowering the correlated error

Ensemble diversity and error correlation: implementation

1. measure baseline bias and variance
2. create a controlled diversity source
3. estimate prediction correlation
4. aggregate consistently
5. verify the variance floor

Check the result

Check base-error correlation and ensemble variance on training and held-out examples.

Common mistake

adding nearly identical learners increases cost without meaningfully lowering the correlated error

Ensemble diversity and error correlation: common mistakes

Example

unstable decision trees trained on nearby samples

Explain the calculation and check base-error correlation and ensemble variance, using a small example.

Common mistake

adding nearly identical learners increases cost without meaningfully lowering the correlated error

What to do

Increase diversity or change the base learner before increasing ensemble size further.

Ensemble diversity and error correlation: check your understanding

Question

When does averaging many predictors reduce generalization error?

Answer

Averaging reduces the independent part of variance but leaves common error untouched.

Decision. Increase diversity or change the base learner before increasing ensemble size further.

How does sampling with replacement create useful training-set variation?



Repeated observations create a perturbed empirical distribution while leaving a predictable out-of-bag subset.

Question

How does sampling with replacement create useful training-set variation?

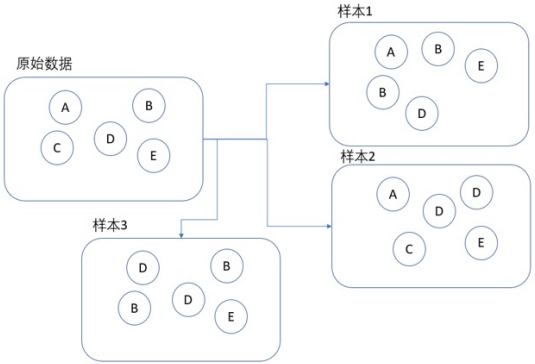
Example

resampling patients with repeated visits

Main idea

Repeated observations create a perturbed empirical distribution while leaving a predictable out-of-bag subset.

Bootstrap sampling uses replacement



Sampling with replacement creates repeated and omitted examples.

Bootstrap sampling: model and variables

Model and assumptions

A bootstrap sample draws N observations with replacement from an original set of N observations.

$$\Pr(i \notin \mathcal{D}^{(b)}) = \left(1 - \frac{1}{N}\right)^{N} \longrightarrow e^{-1} \approx 0.368$$



Symbol	Meaning	Status
$\mathcal{D}^{(b)}$	bootstrap dataset	one resampled training set
N	original sample size	also the draw count
OOB	not selected observations	about 36.8% per tree

Bootstrap sampling: worked example

Quantity	Value	Meaning
original rows	$1, 2, 3, 4, 5$	five observations
bootstrap	$1, 2, 2, 4, 4$	duplicates allowed
OOB	$3, 5$	local validation rows

What the result means

Bootstrap the independent sampling unit, not automatically the table row.

The choice rule for Bootstrap sampling

Main idea

Repeated observations create a perturbed empirical distribution while leaving a predictable out-of-bag subset.

What to check

unique-row fraction and OOB coverage

Limit

resampling individual rows breaks dependence when observations belong to the same patient, site, or time block

Bootstrap sampling: implementation

1. draw exactly N rows with replacement
2. retain row identifiers
3. record OOB membership
4. fit one unstable learner
5. repeat independently

Check the result

Check unique-row fraction and OOB coverage on training and held-out examples.

Common mistake

resampling individual rows breaks dependence when observations belong to the same patient, site, or time block

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Bootstrap sampling: common mistakes

Example

resampling patients with repeated visits

Explain the calculation and check unique-row fraction and OOB coverage, using a small example.

Common mistake

resampling individual rows breaks dependence when observations belong to the same patient, site, or time block

What to do

Bootstrap the independent sampling unit, not automatically the table row.

Bootstrap sampling: check your understanding

Question

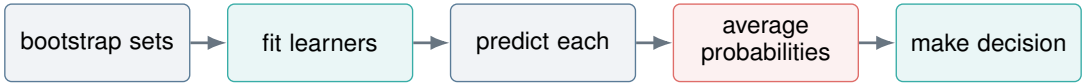
How does sampling with replacement create useful training-set variation?

Answer

Repeated observations create a perturbed empirical distribution while leaving a predictable out-of-bag subset.

Decision. Bootstrap the independent sampling unit, not automatically the table row.

How should bootstrap learners be combined for classification and regression?



Averaging stabilizes a high-variance learner without deliberately changing its expected prediction.

Question

How should bootstrap learners be combined for classification and regression?

Example

bagging deep decision trees for a binary risk score

Main idea

Averaging stabilizes a high-variance learner without deliberately changing its expected prediction.

Averaging reduces variance when errors differ

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Bagging可降低模型方差

- 在Bagging中，M次预测结果的均值$f(\boldsymbol{x})$的方差比用原始训练样本单次训练的模型的预测结果$f_m(\boldsymbol{x})$的方差小，二者的期望相同
  - Bagging可以降低模型方差
  - Bagging不改变模型偏差
- Bagging中每个模型不完全独立（训练样本有一部分相同），方差的减少没那么多，但也会减少
  - 若f_m之间的相关性为ρ，则$f(\boldsymbol{x})$的方差为：$\rho\times\sigma^2+(1-\rho)\times\frac{\sigma^2}{M}$
  - 随机森林相比标准的bagging，对样本和特征进行抽样，进一步减少树之间的相关性

推荐阅读：
1.《为什么说bagging是减少variance，而boosting是减少bias?》
2.使用sklearn进行集成学习——理论https://www.cnblogs.com/jasonfreak/p/5657196.html

Bagging prediction and variance reduction: model and variables

Model and assumptions

Bagging fits one base learner per bootstrap sample and combines predictions by probability averaging or numeric averaging.

$$\widehat{f}_{\mathrm{bag}}(x) = \frac{1}{M}\sum_{m=1}^{M}\widehat{f}^{(m)}(x)$$

Symbol	Meaning	Status
$\widehat{f}^{(m)}$	base prediction	trained on bootstrap m
M	ensemble size	controls Monte Carlo stability
aggregation	mean or class probability	must match the task

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Bagging prediction and variance reduction: worked example

Quantity	Value	Meaning
tree probabilities	$0.2, 0.8, 0.7$	class 1
mean probability	0.567	ensemble score
threshold	0.5	predict class 1

What the result means

Average calibrated class probabilities unless the deployment rule explicitly requires hard votes.

Bagging prediction and variance reduction: checking the result

Main idea

Averaging stabilizes a high-variance learner without deliberately changing its expected prediction.

What to check

ensemble variance and probability calibration

Limit

hard majority voting discards probability strength and can worsen calibration under class imbalance

What to record when running Bagging prediction and variance reduction

```
from sklearn.ensemble import BaggingClassifier
from sklearn.tree import DecisionTreeClassifier
bag = BaggingClassifier(DecisionTreeClassifier(max_depth=None), n_estimators=80, max_samples=0.8, bootstrap=True,
     random_state=7, n_jobs=-1)
bag.fit(X_train, y_train)
print('Bagging accuracy:', round(bag.score(X_test, y_test), 3))
```

Evidence and failure check

Check ensemble variance and probability calibration with the fitted model and data split; reject the run if hard majority voting discards probability strength and can worsen calibration under class imbalance.

The warning sign for Bagging prediction and variance reduction

Example

bagging deep decision trees for a binary risk score

Explain the calculation and check ensemble variance and probability calibration, using a small example.

Common mistake

hard majority voting discards probability strength and can worsen calibration under class imbalance

What to do

Average calibrated class probabilities unless the deployment rule explicitly requires hard votes.

Explain Bagging prediction and variance reduction without naming the algorithm

Question

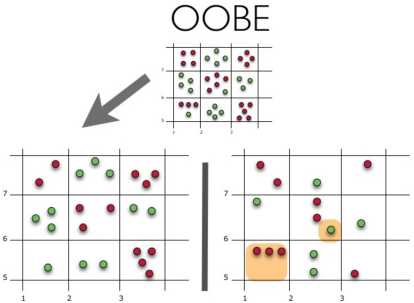
How should bootstrap learners be combined for classification and regression?

Answer

Averaging stabilizes a high-variance learner without deliberately changing its expected prediction.

Decision. Average calibrated class probabilities unless the deployment rule explicitly requires hard votes.

Can each observation be evaluated by trees that did not train on it?



Question

Can each observation be evaluated by trees that did not train on it?

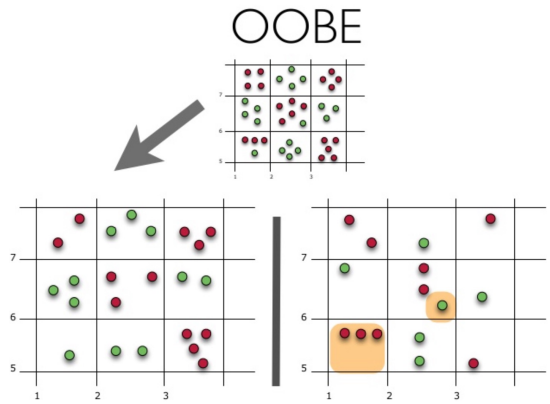
Example

estimating forest error without a separate tuning fold

Main idea

OOB estimates reuse training data efficiently while preserving a per-tree exclusion boundary.

Out-of-bag predictions use trees that missed a sample



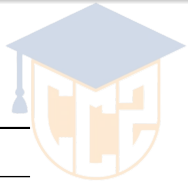
For each observation, use only trees that omitted it from their bootstrap samples.

How Out-of-bag estimation changes the representation

Model and assumptions

An out-of-bag prediction aggregates only the bootstrap learners for which that observation was not selected.

$$\widehat{f}_{\mathrm{OOB}}(x_i) = \frac{1}{|\mathcal{M}_i|} \sum_{m \in \mathcal{M}_i} \widehat{f}^{(m)}(x_i)$$



Symbol	Meaning	Status
$\mathcal{M}_i$	trees excluding observation i	local evaluation ensemble
coverage	number of OOB trees	grows with M
OOB error	loss on OOB predictions	internal generalization estimate

Check the direction of change in Out-of-bag estimation

Quantity	Value	Meaning
trees	100	ensemble
expected OOB trees	36.8	per observation
minimum coverage	20	flag if lower

What the result means

Use grouped validation when the sampling unit is not an independent row.

When Out-of-bag estimation earns a place in the pipeline

Main idea

OOB estimates reuse training data efficiently while preserving a per-tree exclusion boundary.

What to check

OOB error, coverage, and holdout agreement

Limit

OOB evaluation is optimistic when dependent rows from one entity appear in both in-bag and OOB sets

Out-of-bag estimation: implementation

1. store bootstrap membership
2. aggregate only excluding trees
3. check OOB counts
4. compute the target metric
5. compare once with a grouped holdout

Check the result

Check OOB error, coverage, and holdout agreement on training and held-out examples.

Common mistake

OOB evaluation is optimistic when dependent rows from one entity appear in both in-bag and OOB sets

When the objective of Out-of-bag estimation is not enough

Example

estimating forest error without a separate tuning fold

Explain the calculation and check OOB error, coverage, and holdout agreement, using a small example.

Common mistake

OOB evaluation is optimistic when dependent rows from one entity appear in both in-bag and OOB sets

What to do

Use grouped validation when the sampling unit is not an independent row.

A final decision on Out-of-bag estimation

Question

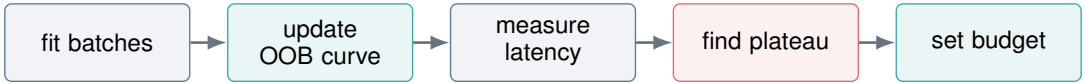
Can each observation be evaluated by trees that did not train on it?

Answer

OOB estimates reuse training data efficiently while preserving a per-tree exclusion boundary.

Decision. Use grouped validation when the sampling unit is not an independent row.

When have additional base learners stopped buying useful stability?



Test performance usually plateaus rather than deteriorates, while memory and latency continue to grow.

Question

When have additional base learners stopped buying useful stability?

Example

setting the number of trees for an online service

Main idea

Test performance usually plateaus rather than deteriorates, while memory and latency continue to grow.

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Add trees until validation performance and runtime level off

- More trees reduce random variation in the ensemble prediction.
- Trees with strongly correlated errors offer less improvement.
- Prediction time and memory usually grow with the number of trees.
- Use OOB or validation curves to choose a practical ensemble size.

The variables that make Ensemble size and parallel execution work

Model and assumptions

Ensemble size controls Monte Carlo variation and compute cost, not the structural complexity of an individual tree.

$$\mathrm{SE}\left(\frac{1}{M}\sum_{m} f_m(x)\right) \propto M^{-1/2} \quad \text{for weakly dependent learners}$$

Symbol	Meaning	Status
M	number of learners	compute and stability
plateau	small marginal gain	stopping criterion
latency	inference cost	deployment constraint

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Read the numbers produced by Ensemble size and parallel execution

Quantity	Value	Meaning
trees	50	OOB 0.842
trees	200	OOB 0.861
trees	800	OOB 0.862; 4× latency

What the result means

Stop after the validation curve stabilizes within the deployment budget.

Three conditions for trusting Ensemble size and parallel execution

Main idea

Test performance usually plateaus rather than deteriorates, while memory and latency continue to grow.

What to check

marginal accuracy gain per unit latency

Limit

selecting M by accuracy alone can create an unnecessarily slow predictor with no practical gain

An implementation checklist for Ensemble size and parallel execution

1. grow the ensemble in batches
2. record OOB and holdout curves
3. measure memory and latency
4. locate a stable plateau
5. retain a compute margin

Check the result

Check marginal accuracy gain per unit latency on training and held-out examples.

Common mistake

selecting M by accuracy alone can create an unnecessarily slow predictor with no practical gain

Reject Ensemble size and parallel execution under this condition

Example

setting the number of trees for an online service

Explain the calculation and check marginal accuracy gain per unit latency, using a small example.

Common mistake

selecting M by accuracy alone can create an unnecessarily slow predictor with no practical gain

What to do

Stop after the validation curve stabilizes within the deployment budget.

Ensemble size and parallel execution: check your understanding

Question

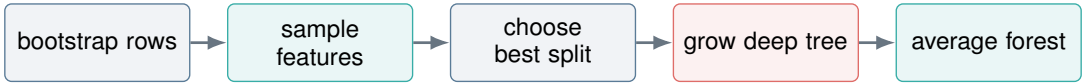
When have additional base learners stopped buying useful stability?

Answer

Test performance usually plateaus rather than deteriorates, while memory and latency continue to grow.

Decision. Stop after the validation curve stabilizes within the deployment budget.

Why does randomizing candidate features make trees more useful together?



Feature subsampling prevents a dominant predictor from forcing the same top split in every tree.

Question

Why does randomizing candidate features make trees more useful together?

Example

many correlated laboratory features

Main idea

Feature subsampling prevents a dominant predictor from forcing the same top split in every tree.

Random feature subsets make trees less alike

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

随机森林（Random Forest）

- 由于只是训练数据有一些不同，对决策树算法进行Bagging得到的多棵树高度相关，因此带来的方差减少有限。
- 随机森林通过
  - 随机选择一部分特征
  - 随机选择一部分样本
- 降低树的相关性

- 随机森林在很多应用案例上被证明有效，但牺牲了可解释性
  - 森林：多棵树
  - 随机：对样本和特征进行随机抽取

Random feature selection in forests: model and variables

Model and assumptions

A random forest combines bootstrap trees with a random subset of candidate features at every split.

$$j^* = \arg\max_{j \in \mathcal{S}_t,\ s} \Delta I(j, s), \qquad |\mathcal{S}_t| = m_{\mathrm{try}} < d$$

Symbol	Meaning	Status
$\mathcal{S}_t$	candidate features at node t	fresh random subset
m_{try}	subset size	correlation-strength tradeoff
ΔI	impurity reduction	split criterion

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Random feature selection in forests: worked example

Quantity	Value	Meaning
features	$d = 100$	input
$m_{\mathrm{try}} = 10$	classification default scale	decorrelates trees
$m_{\mathrm{try}} = 100$	bagged trees	stronger correlation

What the result means

Tune feature subsampling by the joint strength-correlation tradeoff.

A decision card for Random feature selection in forests

Main idea

Feature subsampling prevents a dominant predictor from forcing the same top split in every tree.

What to check

OOB score and pairwise tree correlation

Limit

too few candidate features weaken every split; too many make trees nearly identical

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Reproduce the result of Random feature selection in forests

```
from sklearn.ensemble import RandomForestClassifier
rf = RandomForestClassifier(n_estimators=250, max_features='sqrt', min_samples_leaf=2, oob_score=True, random_state
     =7, n_jobs=-1)
rf.fit(X_train, y_train)
print('RF OOB/test:', round(rf.oob_score_, 3), round(rf.score(X_test, y_test), 3))
```


Evidence and failure check

Check OOB score and pairwise tree correlation with the fitted model and data split; reject the run if too few candidate features weaken every split; too many make trees nearly identical.

A stress test for Random feature selection in forests

Example

many correlated laboratory features

Explain the calculation and check OOB score and pairwise tree correlation, using a small example.

Common mistake

too few candidate features weaken every split; too many make trees nearly identical

What to do

Tune feature subsampling by the joint strength-correlation tradeoff.

Random feature selection in forests: check your understanding

Question

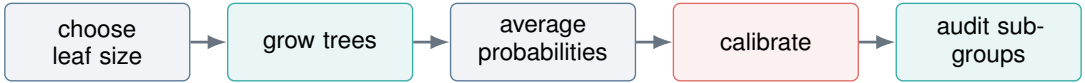
Why does randomizing candidate features make trees more useful together?

Answer

Feature subsampling prevents a dominant predictor from forcing the same top split in every tree.

Decision. Tune feature subsampling by the joint strength-correlation tradeoff.

Should a random-forest tree be fully grown or locally regularized?



Averaging reduces variance, but tiny leaves can still produce unstable and poorly calibrated probabilities.

Question

Should a random-forest tree be fully grown or locally regularized?

Example

risk prediction where probability drives intervention

Main idea

Averaging reduces variance, but tiny leaves can still produce unstable and poorly calibrated probabilities.

Depth and leaf size change how detailed a tree can become

- A deeper tree creates more local regions.
- A larger minimum leaf size prevents splits supported by very few observations.
- A pure leaf with two observations gives a fragile probability estimate.
- Tune complexity on held-out data; averaging trees does not remove all bias.

Tree depth, leaf size, and probability quality: model and variables

Model and assumptions

Forest regularization controls the resolution of each tree through depth, minimum leaf size, and split thresholds.

$$\widehat{p}_m(c \mid x) = \frac{n_{m,c}(\ell(x)) + \alpha}{n_m(\ell(x)) + K\alpha}$$

Symbol	Meaning	Status
$\ell(x)$	terminal leaf	depends on tree path
$n_{m,c}$	class count in leaf	local evidence
α	probability smoothing	prevents extreme estimates

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Limit tree depth and the size of leaves

- A deep tree can fit small, noisy groups of training examples.
- A larger minimum leaf size makes each leaf estimate use more observations.
- Leaf class probability is the class fraction among its training examples.
- Choose these settings using validation predictions, including probability quality.

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Tree depth, leaf size, and probability quality: worked example

Quantity	Value	Meaning
leaf size	1	high resolution; extreme votes
leaf size	10	smoother probabilities
validation	same AUROC	better log loss at size 10

What the result means

Tune and calibrate for the decision metric, not only for ranking accuracy.

Tree depth, leaf size, and probability quality: checking the result

Main idea

Averaging reduces variance, but tiny leaves can still produce unstable and poorly calibrated probabilities.

What to check

log loss, calibration error, and rare-class recall

Limit

a forest can rank cases well while returning overconfident probabilities from tiny leaves

Tree depth, leaf size, and probability quality: implementation

1. tune leaf size before depth
2. compare discrimination and log loss
3. inspect probability histograms
4. calibrate on a separate fold
5. check rare-class recall

Check the result

Check log loss, calibration error, and rare-class recall on training and held-out examples.

Common mistake

a forest can rank cases well while returning overconfident probabilities from tiny leaves

What breaks first in Tree depth, leaf size, and probability quality

Example

risk prediction where probability drives intervention

Explain the calculation and check log loss, calibration error, and rare-class recall, using a small example.

Common mistake

a forest can rank cases well while returning overconfident probabilities from tiny leaves

What to do

Tune and calibrate for the decision metric, not only for ranking accuracy.

Tree depth, leaf size, and probability quality: check your understanding

Question

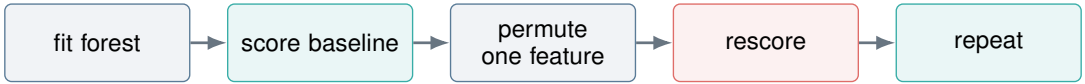
Should a random-forest tree be fully grown or locally regularized?

Answer

Averaging reduces variance, but tiny leaves can still produce unstable and poorly calibrated probabilities.

Decision. Tune and calibrate for the decision metric, not only for ranking accuracy.

Which importance estimate answers the question we actually care about?



Correlated predictors share or substitute importance, so neither score is a causal effect.

Question

Which importance estimate answers the question we actually care about?

Example

explaining a forest with redundant biomarkers

Main idea

Correlated predictors share or substitute importance, so neither score is a causal effect.

Feature importance depends on how it is measured

- Impurity importance sums the training impurity reduction credited to a feature.
- It can favor features with many possible split points.
- Permutation importance measures a held-out score change after shuffling one column.
- Correlated columns can share or hide importance; neither measure proves causation.



Feature importance and correlated predictors: model and variables

Model and assumptions

Impurity importance measures accumulated split improvement; permutation importance measures performance loss after disrupting a feature.

$$I_j^{\mathrm{perm}} = \mathbb{E}\Big[L(y, \widehat{f}(X_{\pi(j)})) - L(y, \widehat{f}(X))\Big]$$

Symbol	Meaning	Status
$X_{\pi(j)}$	feature j permuted	breaks its association
loss increase	predictive importance	depends on metric
correlation	substitution path	changes attribution

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Feature importance and correlated predictors: worked example

Quantity	Value	Meaning
feature A	correlated with B	substitutable
permute A	small loss	B compensates
permute A and B	large loss	group is important

What the result means

Use held-out permutation importance and group correlated features when interpretation matters.

Accept or reject Feature importance and correlated predictors

Main idea

Correlated predictors share or substitute importance, so neither score is a causal effect.

What to check

repeated validation permutation loss

Limit

impurity importance favors continuous or high-cardinality variables and can be read as causal evidence

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Feature importance and correlated predictors: implementation

1. compute importance on validation data
2. repeat permutations
3. group strongly correlated features
4. compare impurity and permutation ranks
5. state the predictive question

Check the result

Check repeated validation permutation loss on training and held-out examples.

Common mistake

impurity importance favors continuous or high-cardinality variables and can be read as causal evidence

The hidden assumption in Feature importance and correlated predictors

Example

explaining a forest with redundant biomarkers

Explain the calculation and check repeated validation permutation loss, using a small example.

Common mistake

impurity importance favors continuous or high-cardinality variables and can be read as causal evidence

What to do

Use held-out permutation importance and group correlated features when interpretation matters.

Feature importance and correlated predictors: check your understanding

Question

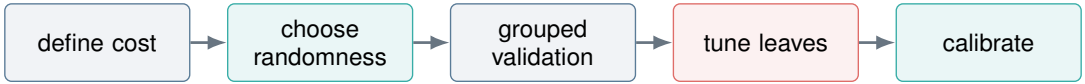
Which importance estimate answers the question we actually care about?

Answer

Correlated predictors share or substitute importance, so neither score is a causal effect.

Decision. Use held-out permutation importance and group correlated features when interpretation matters.

Which additional randomness or weighting matches the operational risk?



Randomness targets variance, while class weighting targets the decision cost; they solve different problems.

Question

Which additional randomness or weighting matches the operational risk?

Example

detecting a two-percent equipment failure rate

Main idea

Randomness targets variance, while class weighting targets the decision cost; they solve different problems.

Random Forests and Extra Trees randomize differently

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Scikit-Learn中的随机森林

- Scikit-Learn中实现了两种包含随机树的森林
  - 随机森林（Random Forests）
  - 极度随机森林（ Extremely Randomized Trees）
- 极度随机森林组合比随机森林更随机
  - 在分裂时，随机森林寻找特征最有判别力的阈值。
  - 极度随机森林中，随机选取每个候选特征的阈值，然后从这些随机选取的阈值中寻找最佳阈值：每个结点上找到每个特征的最佳阈值是决策树生长中最耗时的任务
  - 极度随机森林对方差的减少会更多一些，但偏差可能增大一点点。

Extra Trees, class weights, and robust tuning: model and variables

Model and assumptions

Extremely randomized trees randomize split thresholds; class weights change the loss contribution of rare outcomes.

$$I_w(t) = 1 - \sum_c \left(\frac{\sum_{i \in t} w_{y_i} \mathbf{1}[y_i = c]}{\sum_{i \in t} w_{y_i}} \right)^2$$

Symbol	Meaning	Status
w_{y_i}	class weight	cost or prevalence correction
random threshold	split candidate	additional decorrelation
validation metric	selection target	must match operations

Extra Trees, class weights, and robust tuning: worked example

Quantity	Value	Meaning
positive rate	2%	rare event
class weight	20	raises miss cost
accuracy	98%	uninformative baseline

What the result means

Tune against the operational cost and show the threshold-dependent tradeoff.

Separate a good fit from a good decision in Extra Trees, class weights, and robust tuning

Main idea

Randomness targets variance, while class weighting targets the decision cost; they solve different problems.

What to check

precision-recall curve, expected cost, and calibration

Limit

optimizing raw accuracy on a rare-event task can select a model that predicts no positives

Extra Trees, class weights, and robust tuning: implementation

```
from sklearn.ensemble import ExtraTreesClassifier
et = ExtraTreesClassifier(n_estimators=250, max_features='sqrt', min_samples_leaf=2, class_weight='balanced',
     random_state=7, n_jobs=-1)
et.fit(X_train, y_train)
print('Extra Trees accuracy:', round(et.score(X_test, y_test), 3))
```


Evidence and failure check

Check precision-recall curve, expected cost, and calibration with the fitted model and data split; reject the run if optimizing raw accuracy on a rare-event task can select a model that predicts no positives.

Do not trust Extra Trees, class weights, and robust tuning before this check

Example

detecting a two-percent equipment failure rate

Explain the calculation and check precision-recall curve, expected cost, and calibration, using a small example.

Common mistake

optimizing raw accuracy on a rare-event task can select a model that predicts no positives

What to do

Tune against the operational cost and show the threshold-dependent tradeoff.

Extra Trees, class weights, and robust tuning: check your understanding

Question

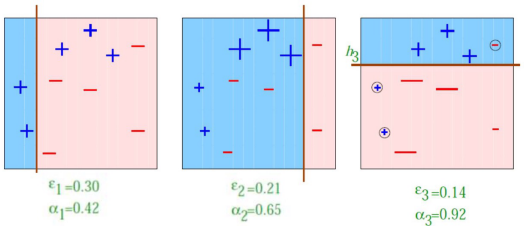
Which additional randomness or weighting matches the operational risk?

Answer

Randomness targets variance, while class weighting targets the decision cost; they solve different problems.

Decision. Tune against the operational cost and show the threshold-dependent tradeoff.

When should independent averaging give way to sequential error correction?



Question

When should independent averaging give way to sequential error correction?

Example

choosing the next ensemble family after a tree baseline

Main idea

The choice follows the error profile: unstable strong learners favor bagging, while weak biased learners can benefit from boosting.

Boosting trains the next learner on current mistakes

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

怎样得到互补的学习器？

- 在不同的训练集上训练学习器。
- 怎么得到不同的训练集?
  - 对原始训练集重采样
  - 改变训练集的标签（基于梯度的提升法）
  - 对原始训练集重新加权（AdaBoost）
    - 在实际操作中，改变目标函数即可：

$$(\boldsymbol{x}_1, y_1, w_1)$$
$$\cdots$$
$$(\boldsymbol{x}_N, y_N, w_N)$$

$$J(f,\lambda) = \sum_{i=1}^{N} L(y_i, f(\mathbf{x}_i)) + \lambda R(f)$$

$$J(f,\lambda) = \sum_{i=1}^{N} w_i L(y_i, f(\mathbf{x}_i)) + \lambda R(f)$$

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Bagging versus boosting: model and variables

Model and assumptions

Bagging fits peers in parallel to reduce variance; boosting fits learners in sequence to reduce residual error or exponential loss.

$$F_M(x) = \sum_{m=1}^{M} \alpha_m h_m(x)$$

Symbol	Meaning	Status
h_m	weak learner	added at round m
α_m	learner weight	sequential contribution
residual or weight	next-round signal	depends on boosting family

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Bagging versus boosting: worked example

Quantity	Value	Meaning
bagging	parallel peers	variance reduction
boosting	sequential corrections	bias and loss reduction
shared idea	aggregate models	validate complexity

What the result means

Use boosting when sequential corrections are stable; retain bagging when variance is the dominant problem.

Bagging versus boosting: checking the result

Main idea

The choice follows the error profile: unstable strong learners favor bagging, while weak biased learners can benefit from boosting.

What to check

training-validation curves by ensemble round

Limit

boosting noisy labels with overly deep learners can focus capacity on irreducible errors

Bagging versus boosting: implementation

1. diagnose the baseline error
2. choose averaging or correction
3. set weak-learner complexity
4. track validation performance by round
5. stop before unnecessary complexity

Check the result

Check training-validation curves by ensemble round on training and held-out examples.

Common mistake

boosting noisy labels with overly deep learners can focus capacity on irreducible errors

Bagging versus boosting: limitations

Example

choosing the next ensemble family after a tree baseline

Explain the calculation and check training-validation curves by ensemble round, using a small example.

Common mistake

boosting noisy labels with overly deep learners can focus capacity on irreducible errors

What to do

Use boosting when sequential corrections are stable; retain bagging when variance is the dominant problem.

Bagging versus boosting: check your understanding

Question

When should independent averaging give way to sequential error correction?

Answer

The choice follows the error profile: unstable strong learners favor bagging, while weak biased learners can benefit from boosting.

Decision. Use boosting when sequential corrections are stable; retain bagging when variance is the dominant problem.

OOB and cross-validation answer different questions

Decision

Use out-of-bag predictions for an efficient internal estimate and cross-validation for the full model-selection procedure.

Evidence

Compare the two estimates with identical metrics and confidence intervals.

Stop condition

Investigate data grouping, temporal order, or leakage when the estimates disagree systematically.

Choose a forest under a deployment budget

Decision

Treat accuracy, probability quality, latency, memory, and update frequency as one decision.

Evidence

Profile the complete preprocessing-and-prediction path at realistic batch sizes.

Stop condition

Do not trade a tiny validation gain for a model that misses the service limit or produces uncalibrated probabilities.

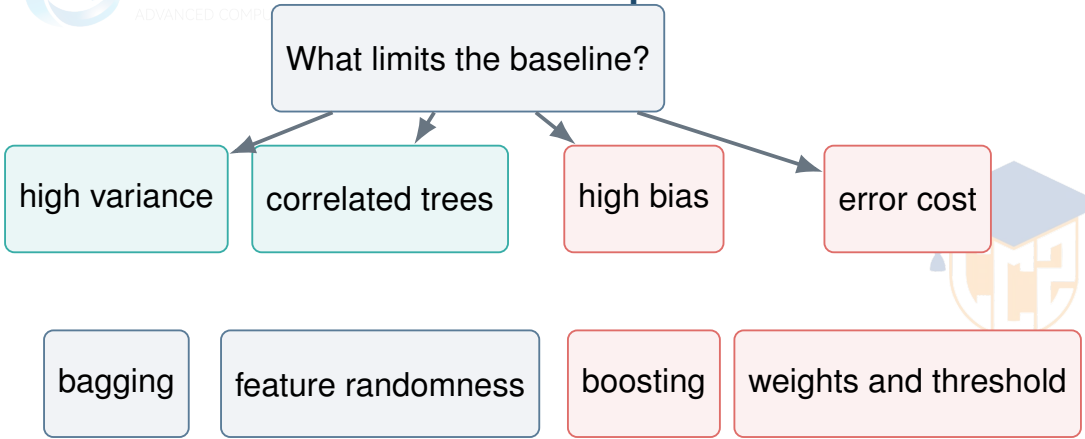
先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Compare ensemble families by their source of improvement

Family	Main mechanism	Best first diagnostic
bagging	row resampling and averaging	variance and OOB curve
random forest	row and feature randomness	strength versus correlation
Extra Trees	random thresholds	bias-variance comparison
boosting	sequential corrections	round-wise validation curve



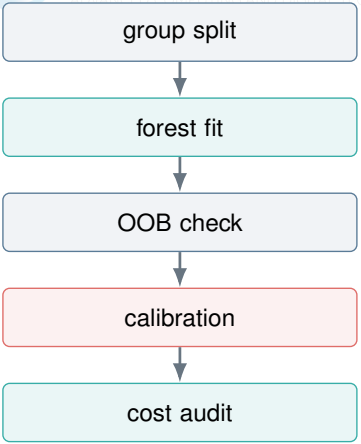
Choose the ensemble from the error profile



Start from the deliverable

Diagnose high variance, correlated trees, class-cost mismatch, or persistent bias before choosing the mechanism.

A reliable forest workflow



OOB is useful evidence, not a waiver from dependency-aware validation.

1. split by the independent entity
2. fit preprocessing inside folds
3. track OOB and grouped validation
4. tune leaf size and feature subsampling
5. calibrate on held-out predictions
6. measure latency at the chosen threshold

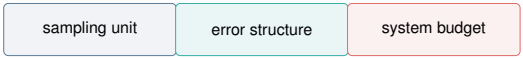
Separate accuracy gain from diversity gain

$$\Delta_M = \text{loss}(f_1) - \text{loss}\left(M^{-1}\sum_{m=1}^{M} f_m\right)$$

Observation	Likely limit	Next action
single trees strong; ensemble unstable	too few trees	increase M
trees strong; highly correlated	weak diversity	lower feature fraction
train and test both poor	high bias	change features or boost

Stress-test the forest before deployment

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER



- repeat seeds and grouped splits
- compare OOB with external validation
- sweep leaf size and feature fraction
- audit classwise calibration
- permute correlated feature groups
- measure memory and tail latency

Six takeaways to retain

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

1. bagging primarily reduces variance
2. bootstrap the independent unit
3. OOB predictions exclude the current observation
4. random features trade tree strength for diversity
5. importance is predictive rather than causal
6. boosting is sequential correction, not parallel averaging

Exit ticket: choose an ensemble under a budget

Prompt

Choose bagging, a random forest, Extra Trees, or a boosting pilot. State the diagnosed error, one diversity control, one validation metric, and one deployment constraint.

Case

A deep tree is accurate on training data but unstable across hospital sites. The service allows 20 ms per prediction and requires calibrated probabilities for a rare outcome.